Activity = 2

① Given $G_x = 24$

$G_y = 18$

1) Gradient magnitude $= \sqrt{G_x^2 + G_y^2}$

$= \sqrt{(24)^2 + (18)^2}$

$= \sqrt{576 + 324}$

$= \sqrt{900}$

$= 30$

2) Gradient direction $= \theta = \tan^{-1}(G_y/G_x)$

$\theta = \tan^{-1}\left(\frac{18}{24}\right)$

$\theta = \tan^{-1}(0.75)$

$\theta = 36.87^\circ$

②

Given,

Pixel	Magnitude
P_1	25
P_2	50
P_3	90
P_4	60
P_5	30

since the gradient direction is horizontal.

Compare each pixel with its left and right neighbors.

P1 → Border pixel → supressed (0)

P2 → 50 ⇒ compare with 25 & 90

Not maximum so (0)

P3 → 90 ⇒ Compare with 50 & 60

It is maximum so (90)

P4 → 60 ⇒ compare with 90 & 30

Not maximum so (0)

P5 → 50 ⇒ border so (0) supressed

pixel	P1	P2	P3	P4	P5
output	0	0	90	0	0

(3)

Given,

High Threshold = 100

low Threshold = 50

Pixels & Magnitude		classification	
135	P1	P1 = 135 > 100	strong edge
90	P2	P2 = 90 < 100	weak
45	P3	P3 = 45 < 100	discard
120	P4	P4 = 120 > 100	strong
70	P5	P5 = 70 < 100	weak
20	P6	P6 = 20 < 100	discard

Strong edges : P1 , P4

weak edges : P2 , P5

Discard edges : P3 , P6

(4) Given,

Pixel	Magnitude
P1	135
P2	90
P3	40
P4	115
P5	65
P6	25
P7	105

HT = 100

LT = 50

Given . P2 is connected to P1

P5 is not connected to any strong edges.

strong edges : P1 , P4 , P7

weak edges : P2 , P5

Non edge : P3 , P6

Hysteresis :

P2 is connected to strong edge P1 so keep as edge.

P5 is not connected to any edge so Remove

. final edge pixels after hysteresis

P1, P2, P4, P7